

Mid exam

CPIT250

الترم الأول 2024

22.5

PART A:

Chapter 1: Systems Development in an Organizational Context

Question 1: Fill in the blank from the word bank

Application Software, Tools, Physical design, Implementation, Cutover, Computer-Aided Software Engineering (CASE), Waterfall SDLC, Prototyping, Agile Methodologies, eXtreme programming, Information Systems (IS), Maintenance, Logical design

In _____, all functional features of the system chosen for development in .1
.analysis are described independently of any computer platform

is a diagramming tool that enables graphical representation and _____.2
helps prototype how systems “look and feel

is the study of complementary networks of hardware and software _____.3
that people and organizations use to collect, filter, process, create, and
.distribute data

Rational Unified Process (RUP) establishes four phases of development: .4
_____. inception, elaboration, construction, and

:Question 2

What is an agile methodology for software development? Mention the three reasons when agile methodology may be used

PART B:

Chapter 6: Determining System Requirements

Question 1: Fill in the blank from the word bank

Open-ended questions, Business Process Reengineering (BPR), Direct Observation, Closed-ended questions, Studying Business Documents, System prototypes, Joint Application Design (JAD), Computer-Aided Software Engineering (CASE), Individual Interviews

is an iterative development process; rudimentary working version _____ .1
of a system is built and refined to understand system requirements in concrete
.terms

In _____, an analyst can put the interviewee at ease because he/she can .2
.respond in his/her own words using his/her own structure

is one of the requirements determination methodologies and can _____ .3
.cause people to change their normal operating behaviors

The primary purpose of using _____ is to collect system requirements .4
.simultaneously from the key people involved with the system

Question 2

1.List two drawbacks of individual interviews .

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2.List two participants in Joint Application Design (JAD) sessions .

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PART C:

Chapter 7: Analyzing System Process Requirements: DFD

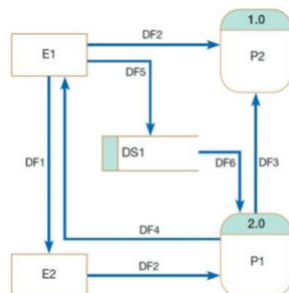
Question 1: Choose the correct answer

One of the following is a data flow diagram which represents a system's major :processes, data flows, and data stores at a high level of detail

- a. Context Diagram
- b. Decomposition
- c. Level-n
- d. Level-0 diagram**

A data flow that appears on a higher level of DFD but not on a lower level is an :example of

- a. Violation of completeness
- b. Violation of consistency**
- c. Gap Analysis
- d. Structuring violation



?which of the following does not follow the rules of data flow diagramming .15

- a. Data flow from source E1 to process P2
- b. Data flow from source E2 to process p1
- c. Data flow from process P1 to process P2
- d. Data flow from source E1 to data store DS1**

? follow the rules of data flow diagramming Which of the following does not .4

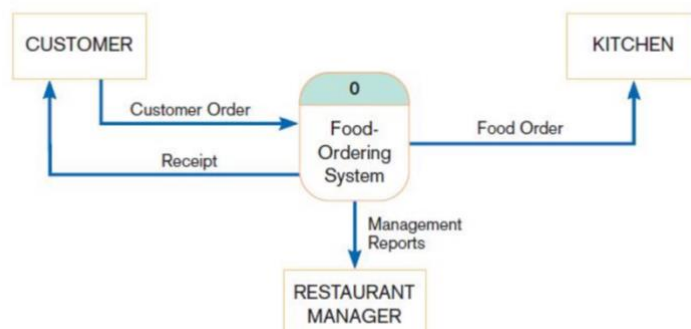
- a. Data flow from data store DS1 to process P1
- b. Data flow from source E1 and process P1 towards process P2 and no data flow out of process P2**
- c. Data flow from process P1 to sink E1
- d. Data flow from source E2 to process p1

Part E:

DFD diagram:

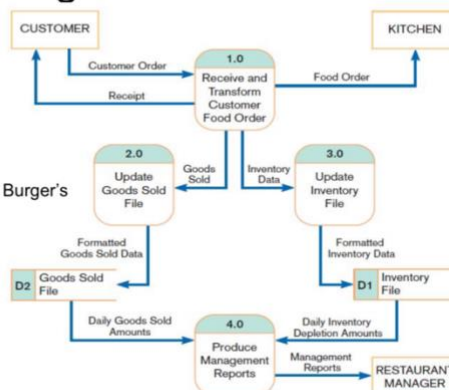
Draw level 0

Context Diagram



Level-0 Diagram

FIGURE 7-5
Level-0 DFD of Hoosier Burger's food-ordering system



PART D:

Chapter 7A: Object-Oriented Analysis and Design: Use Cases (UML)

Question 1: Choose the correct answer

(**True**/False) Use case modeling helps developers understand the functional .1 requirement of the system without worrying about how those requirements will be .implemented

2. [True/**False**] Abstract use case is when a use case is initiated by the user

[True/**False**] The use case that is "extended" represents an association .3 between two use cases where one use case uses the functionality contained in .the other one

(**True**/False) In written use cases, "minimal guarantee" is the outputs that can .4 .be expected if the service attempt failed

Question 2: Use Case Diagramming

Draw a use case model for the following "Student Grade System" describing all possible relationships. Here is a complete list of the requirements for the :system

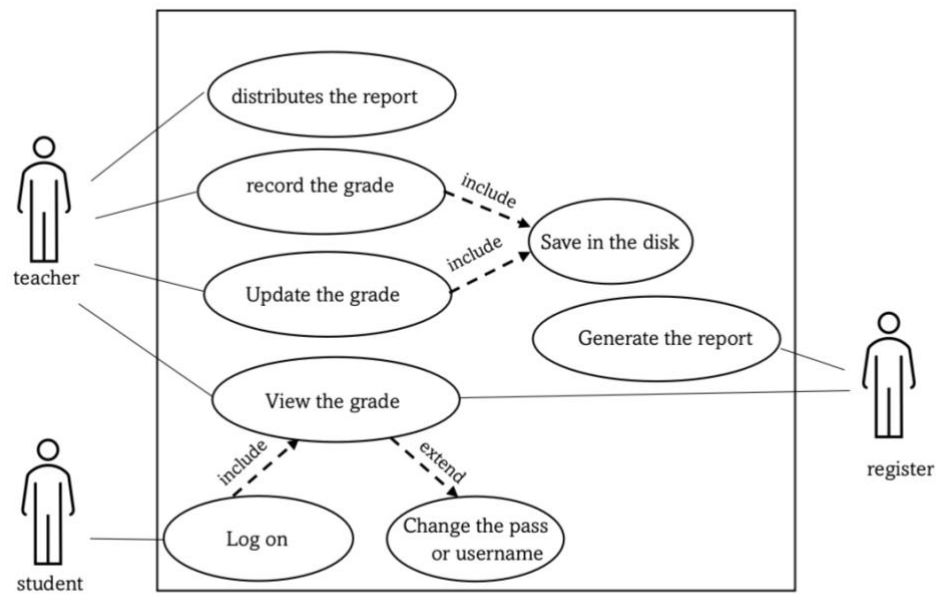
- a teacher can record grades. Whenever grades are recorded, they are also .saved to disk
- a teacher can update grades. Whenever grades are updated, the updated grade .is saved to disk
- a teacher, a registrar, and a student can view grades
- Whenever any of these users view grades, they must always log on to the system. If their log on fails, they must re-authenticate their username and .password

- A registrar can generate reports

- a teacher can distribute reports

.Note: Consider using include and extend use cases

Solve :



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